

JANUARY 2026

FEBRUARY 2026

15

January

26 015-350
3RD WEEK

THURSDAY

Homework 10

(R codes submitted separately)

1) We are given a kernel regression estimator with uniform kernel weights:

$$w_{i,h}(x) = \frac{1(|x_i - x| < h)}{\sum_{i=1}^n 1(|x_i - x| < h)}$$

We need to estimate:

$$\hat{m}(0) = \arg \min_{\theta} \sum_{i=1}^n w_{i,h}(0) (y_i - \theta)^2 \text{ with } h = 0.5$$

This is a weighted least squares problem.
The minimizer of $\sum w_i (y_i - \theta)^2$

is simply: $\hat{\theta} = \sum w_i y_i$

For uniform kernel weights are:

- Equal for all points satisfying: $|x_i - 0| < 0.5$
- Zero otherwise

$$\Rightarrow \hat{m}(0) = \frac{1}{N_0} \sum_{i: |x_i| < 0.5} y_i \text{ where } N_0 = \# \{i: |x_i| < 0.5\}$$

$$\Rightarrow \hat{m}(0) = \frac{1}{N_0} \sum_{i: |x_i| < 0.5} y_i = 0.00785 \text{ (Based on R code)}$$

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January

FRIDAY

016-349
3RD WEEK

26

NOVEMBER 2025

Sun	30	2	9	16	23
Mon		3	10	17	24
Tue		4	11	18	25
Wed		5	12	19	26
Thu		6	13	20	27
Fri		7	14	21	28
Sat	1	8	15	22	29

DECEMBER 2022

Sun		7	14	21
Mon	1	8	15	22
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12)

We need to

08.00

- Write the expression for bias
- Compute it when the true function is linear with slope 1

09.00

Definition of bias: $\text{Bias}(\hat{m}(0)) = E[\hat{m}(0)] - m(0)$

10.00

Substituting $\hat{m}(0)$ from problem 1,

11.00

$$\hat{m}(0) = \frac{1}{N_0} \sum_{i: |x_i| < 0.5} y_i$$

12

We know that $y_i = m(x_i) + \epsilon_i$

13.00

$$\Rightarrow \hat{m}(0) = \frac{1}{N_0} \sum (m(x_i) + \epsilon_i)$$

14.00

Since $E[\epsilon_i] = 0 \Rightarrow E[\hat{m}(0)] = \frac{1}{N_0} \sum_{i: |x_i| < 0.5} m(x_i)$

15.00

$$\Rightarrow \text{Bias} = \frac{1}{N_0} \sum_{i: |x_i| < 0.5} m(x_i) - m(0)$$

16.00

$$\Rightarrow \text{Bias}(\hat{m}(0)) = \frac{1}{N_0} \sum_{i: |x_i| < 0.5} m(x_i) - m(0)$$

17.00

18.00

Given $m(x)$ is linear with slope 1 $\Rightarrow m(x) = x$.

Notes

$$\Rightarrow m(0) = 0$$

$$\Rightarrow \text{Bias}(\hat{m}(0)) = \frac{1}{N_0} \sum_{i: |x_i| < 0.5} x_i = -0.01079$$

(Based on R code)

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January

'26 017-348
3RD WEEK

SATURDAY

13) We need the variance of the estimator, in terms of σ^2 .

We know that, $\hat{m}(0) = \frac{1}{N_0} \sum_{i: |x_i| < 0.5} y_i$

$$\Rightarrow \hat{m}(0) = \frac{1}{N_0} \sum (m(x_i) + \sigma \varepsilon_i) = \frac{1}{N_0} \sum m(x_i) + \frac{\sigma}{N_0} \sum \varepsilon_i$$

We also know that, Variance of a constant = 0
 $E[\varepsilon_i] = 0$; $\text{Var}(\varepsilon_i) = 1$

$$\Rightarrow \text{Var}(\hat{m}(0)) = \text{Var}\left(\frac{\sigma}{N_0} \sum \varepsilon_i\right)$$

$$\Rightarrow \text{Var}\left(\sum \varepsilon_i\right) = N_0 \Rightarrow \text{Var}(\hat{m}(0)) = \frac{\sigma^2}{N_0^2} \cdot N_0$$

$$\Rightarrow \text{Var}(\hat{m}(0)) = \frac{\sigma^2}{N_0} \quad \text{where } N_0 = \# \{i: |x_i| < h\}$$

18 Sunday

14) To estimate σ^2 .

Given

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \hat{m}(x_i))^2$$

We need $\hat{m}(x_i)$ (We need it for all points)

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January

MONDAY

019-346
4TH WEEK 26

NOVEMBER 2025

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DECEMBER 2025

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For each x_i : $\hat{m}(x_i) = \frac{1}{N_i} \sum_{j: |x_j - x_i| < 0.5} y_j$

Then compute: $\hat{\sigma}^2 = \frac{1}{n-1} \sum (y_i - \hat{m}(x_i))^2$

$\hat{\sigma}^2 = 0.04203$ (Based on R code)

11.00

12

15) We need a confidence interval for the predictive value of y at $x=0$ using: $\hat{m}(0)$ from 1) $\hat{\sigma}^2$ from 4)

Normal approximation.

We want: $y(0) = m(0) + \epsilon$

$\text{Var}(y(0)) = \text{Var}(\hat{m}(0)) + \sigma^2$

We know $\text{Var}(\hat{m}(0)) = \frac{\sigma^2}{N_0} \Rightarrow \text{Var}(y(0)) = \sigma^2 \left(1 + \frac{1}{N_0}\right)$

$\Rightarrow \hat{\text{Var}}(y(0)) = \hat{\sigma}^2 \left(1 + \frac{1}{N_0}\right)$

Notes

Using normal approximation,

$\hat{m}(0) \pm 1.96 \cdot \sqrt{\hat{\sigma}^2 \left(1 + \frac{1}{N_0}\right)}$

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95% CI: $[-0.40715, 0.42284]$

(Based on R code)

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2)) We want to split $x \in (-3, 3)$ into two regions at point r :

$$\hat{m}(x) = \begin{cases} m_1 & x \leq r \\ m_2 & x > r \end{cases}$$

For a given r , the optimal m_1 & m_2 that minimizes SSE are simply the means of y in each region:

$$m_1(r) = \frac{\sum_{i: x_i \leq r} y_i}{\# \{i: x_i \leq r\}}, \quad m_2(r) = \frac{\sum_{i: x_i > r} y_i}{\# \{i: x_i > r\}}$$

The total SSE for a given r is:

$$SSE(r) = \sum_{i: x_i \leq r} (y_i - m_1)^2 + \sum_{i: x_i > r} (y_i - m_2)^2$$

So, the algorithm becomes:

- Sort (x_i, y_i) by x_i
- For each candidate split r : Compute: $m_1 = \text{mean}(y_i: x_i \leq r)$
 $m_2 = \text{mean}(y_i: x_i > r)$
- Compute SSE
- Choose r minimizing SSE.

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January

WEDNESDAY

021-344
47TH WEEK '26

NOVEMBER 2025

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DECEMBER 2025

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Using the mid point approach in R code, we get

$$\text{Optimal } r = -0.06695 //$$

$$\text{Minimum SSE} = 10.79163 //$$

2)2) Optimal split point: $r_{opt} = -0.06695$

$$m_1 = \text{mean}(y_i | x_i \leq r_{opt})$$

$$m_2 = \text{mean}(y_i | x_i > r_{opt})$$

$$\text{Final model form: } \hat{m}(x) = \begin{cases} m_1, & x \leq r_{opt} \\ m_2, & x > r_{opt} \end{cases}$$

$$\Rightarrow m(x) = -0.71759 \quad \text{if } x \leq -0.06695 //$$

$$m(x) = 0.66362 \quad \text{if } x > -0.06695 //$$

$$\text{Minimum SSE} = 10.79163 //$$

Notes

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January

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4TH WEEK

22

THURSDAY

2)3) We know $\hat{m}(x) = \begin{cases} m_1, & x \leq r \\ m_2, & x > r \end{cases}$ y corresponding to $x = 0.5$

$$\Rightarrow \hat{m}(0.5) = \begin{cases} m_1, & \text{if } 0.5 \leq r_{\text{opt}} \\ m_2, & \text{if } 0.5 > r_{\text{opt}} \end{cases}$$

As $r_{\text{opt}} = -0.06695$,

$$\Rightarrow \hat{m}(0.5) = m_2 = 0.66362 \quad \text{(Based on R code)}$$

2)4) We need to estimate: $\text{Var}(\hat{m}(0.5))$ using non parametric bootstrap.

For $b=1, \dots, B$.

Sample n observations with replacement.

Fit regression tree (find r_b, m_1^b, m_2^b)

Compute:

$$\hat{m}^{(b)}(0.5)$$

Estimate variance,

$$\hat{\text{Var}} = \frac{1}{B-1} \sum_{b=1}^B \left(\hat{m}^{(b)}(0.5) - \bar{m} \right)^2$$

$$\text{where } \bar{m} = \frac{1}{B} \sum_{b=1}^B \hat{m}^{(b)}(0.5)$$

$$\Rightarrow \hat{\text{Var}}(\hat{m}(0.5)) = 0.12159 \quad \text{(Based on R code)}$$

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FRIDAY

January

023-342
4TH WEEK

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NOVEMBER 2025

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2) 5) Approx 95% CI for $\hat{m}(0.5)$

08.00

$$\Rightarrow \hat{m}(0.5) \sim N(m(0.5), \text{Var}(\hat{m}(0.5)))$$

09.00

$$\text{So CI is: } \hat{m}(0.5) \pm 1.96 \cdot \sqrt{\hat{\text{Var}}}$$

10.00

$$\Rightarrow \hat{m}(0.5) \pm 1.96 \cdot \sqrt{\hat{\text{Var}}(\hat{m}(0.5))}$$

11.00

$$\Rightarrow 95\% \text{ CI for } \hat{m}(0.5) = [-0.01982, 1.34705]$$

13.00

Based on R code

14.00

15.00

16.00

17.00

18.00

Notes